LAB 7

Task 1

Editor - C:\Users\Student\task_1.m

task 01.m | task 02.m | task 03.m | task 05.m | task 06.m | task_1.m | task_2.m | task3_4.m | task_5.m | task_6.m

```
% Task 1 Solution
syms n z

% (a) x[n] = (5*2^n - 3n) u(n)
x_a = (5*2^n - 3*n);
X_a = ztrans(x_a, n, z);
disp('Task 1a Z-transform:'); disp(X_a);

% (b) x[n] = (7*3^n - 4*(-1)^n) u(n)
x_b = (7*3^n - 4*(-1)^n);
X_b = ztrans(x_b, n, z);
disp('Task 1b Z-transform:'); disp(X_b);

% (c) x[n] = (6n + 2*e^(-5n)) u(n)
x_c = (6*n + 2*exp(-5*n));
X_c = ztrans(x_c, n, z);
disp('Task 1c Z-transform:'); disp(X_c);

% (d) x[n] = cos(pi*n) u(n)
x_d = cos(pi*n);
X_d = ztrans(x_d, n, z);
disp('Task 1d Z-transform:'); disp(X_d);

% Symbolic Verification via iztrans
disp('Inverse of 1a:'); disp(iztrans(X_a, z, n));
```

Figure 1: Task 1: Calculated Z-Transforms

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Task 1: Calculated Z-Transforms & Symbolic Verification

(a) $x[n] = (5 \cdot 2^n - 3n)\,u[n] \rightarrow X(z) = (5z)/(z - 2) - (3z)/(z - 1)^2$

(b) $x[n] = (7 \cdot 3^n - 4(-1)^n)\,u[n] \rightarrow X(z) = (7z)/(z - 3) - (4z)/(z + 1)$

(c) $x[n] = (6n + 2e^{-5n})\,u[n] \rightarrow X(z) = (6z)/(z - 1)^2 + (2z)/(z - \exp(-5))$

(d) $x[n] = \cos(\pi n)\,u[n] \rightarrow X(z) = (z(z + 1))/(2z + z^2 + 1)$

Task 2

Editor - C:\Users\Student\task_2.m

task 01.m | task 02.m | task 03.m | task 05.m | task 06.m | task_1.m | task_2.m | task3_4.m | task_5.m

```
% Task 2 MATLAB Code
b1 = [1 1]; a1 = [1 -0.9 0.81];
b2 = [2 5 12]; a2 = [1 2 10];

z1 = roots(b1); p1 = roots(a1);
z2 = roots(b2); p2 = roots(a2);

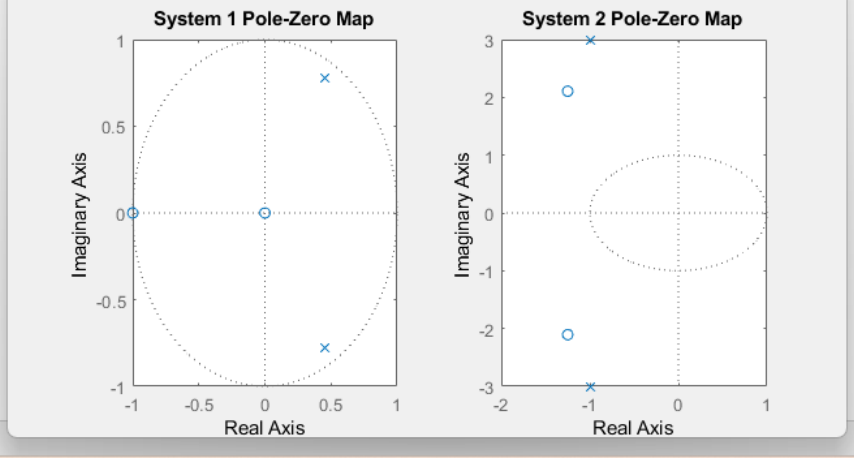
figure;
subplot(1,2,1); pzmap(tf(b1, a1, 1, 'Variable', 'z^-1')); title('System 1 Pole-Zero Map');
subplot(1,2,2); pzmap(tf(b2, a2, 1, 'Variable', 'z^-1')); title('System 2 Pole-Zero Map');
```

Command Wind

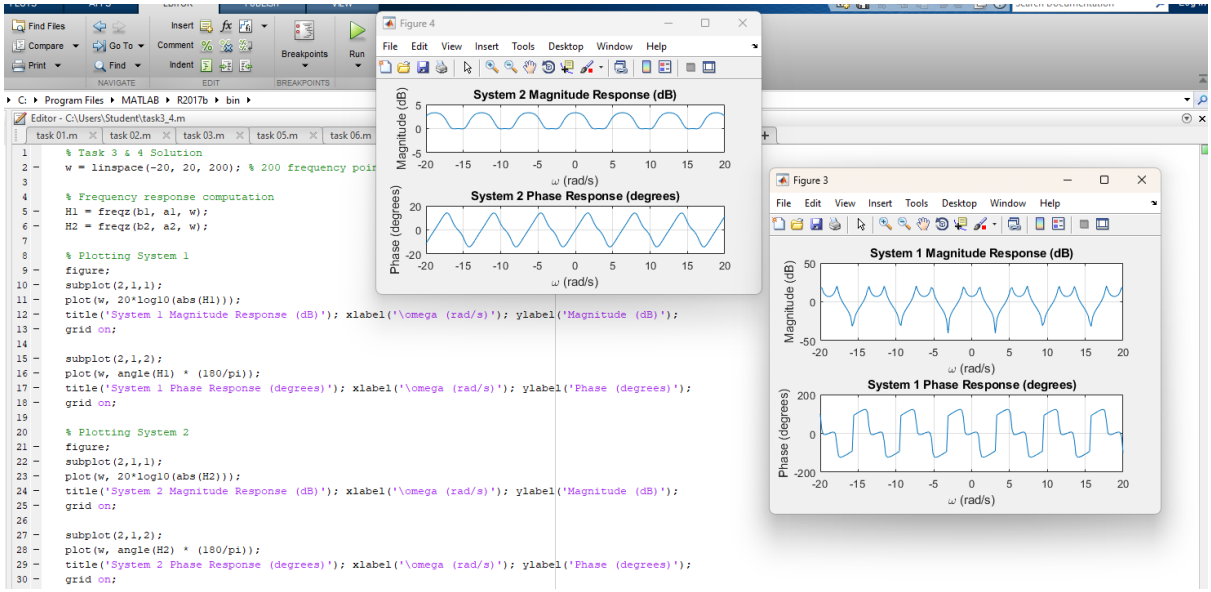
```
>> task_2
>>
```

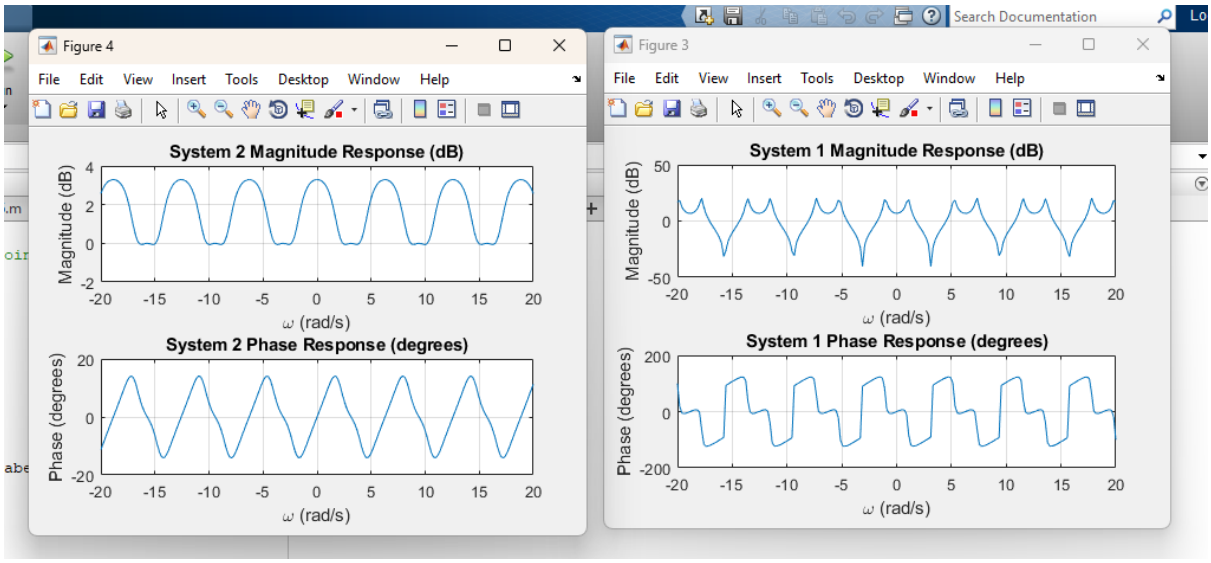
Figure 2

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Task 3 and 4


% Task 3 & 4 Solution
w = linspace(-20, 20, 200); % 200 frequency poir
% Frequency response computation
H1 = freqz(b1, a1, w);
H2 = freqz(b2, a2, w);
% Plotting System 1
figure;
subplot(2,1,1);
plot(w, 20*log10(abs(H1)));
title('System 1 Magnitude Response (dB)'); xlabel('\omega (rad/s)'); ylabel('Magnitude (dB)');
grid on;
subplot(2,1,2);
plot(w, angle(H1) * (180/pi));
title('System 1 Phase Response (degrees)'); xlabel('\omega (rad/s)'); ylabel('Phase (degrees)');
grid on;
% Plotting System 2
figure;
subplot(2,1,1);
plot(w, 20*log10(abs(H2)));
title('System 2 Magnitude Response (dB)'); xlabel('\omega (rad/s)'); ylabel('Magnitude (dB)');
grid on;
subplot(2,1,2);
plot(w, angle(H2) * (180/pi));
title('System 2 Phase Response (degrees)'); xlabel('\omega (rad/s)'); ylabel('Phase (degrees)');
grid on;
System 2 Magnitude Response (dB)
System 2 Phase Response (degrees)
System 1 Magnitude Response (dB)
System 1 Phase Response (degrees)
Magnitude (dB)
Phase (degrees)
ω (rad/s)


Figure 4
Figure 3
File Edit View Insert Tools Desktop Window Help
System 2 Magnitude Response (dB)
System 2 Phase Response (degrees)
System 1 Magnitude Response (dB)
System 1 Phase Response (degrees)
Magnitude (dB)
Phase (degrees)
ω (rad/s)

Task 5

Editor - C:\Users\Student\task_5.m

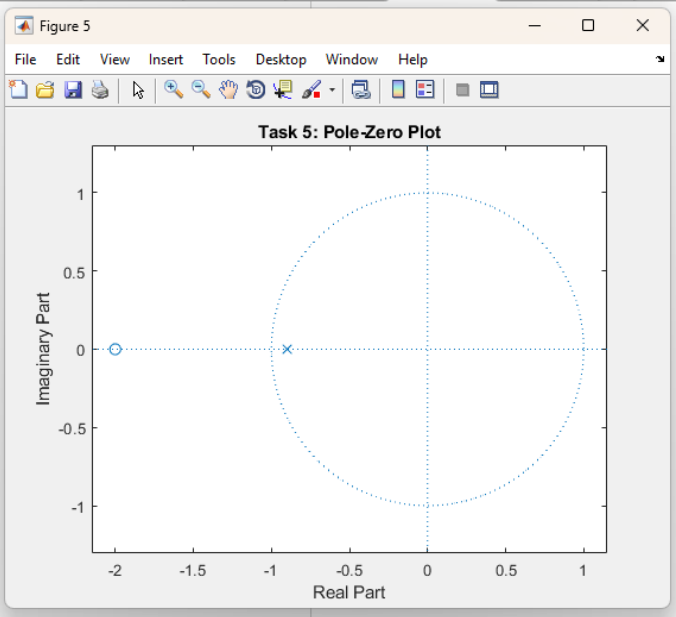
task 01.m | task 02.m | task 03.m | task 05.m | task 06.m | task_1.m | task_2.m | task3_4.m | task_5.m | task_6.m

```
% Task 5 Solution
b = [1 2];    % Numerator coefficients
a = [1 0.9];  % Denominator coefficients

z_pts = roots(b);
p_pts = roots(a);

disp('Zeros:'); disp(z_pts);
disp('Poles:'); disp(p_pts);

% Plot on Z-plane
figure;
zplane(b, a);
title('Task 5: Pole-Zero Plot');
```



Task 6

Editor - C:\Users\Student\task_6.m

task 01.m | task 02.m | task 03.m | task 05.m

```
% Task 6 Solution
num = [1 -3 4 1];
den = [1 -4 1 -0.16];

[r, p, k] = residuez(num, den);

disp('Residues (r):'); disp(r);
disp('Poles (p):'); disp(p);
disp('Direct term (k):'); disp(k);
```

Command Window

```
>> task_6
Residues (r):
   0.5383 + 0.0000i
   3.3559 + 5.7659i
   3.3559 - 5.7659i

Poles (p):
   3.7443 + 0.0000i
   0.1278 + 0.1625i
   0.1278 - 0.1625i

Direct term (k):
   -6.2500

>>
```